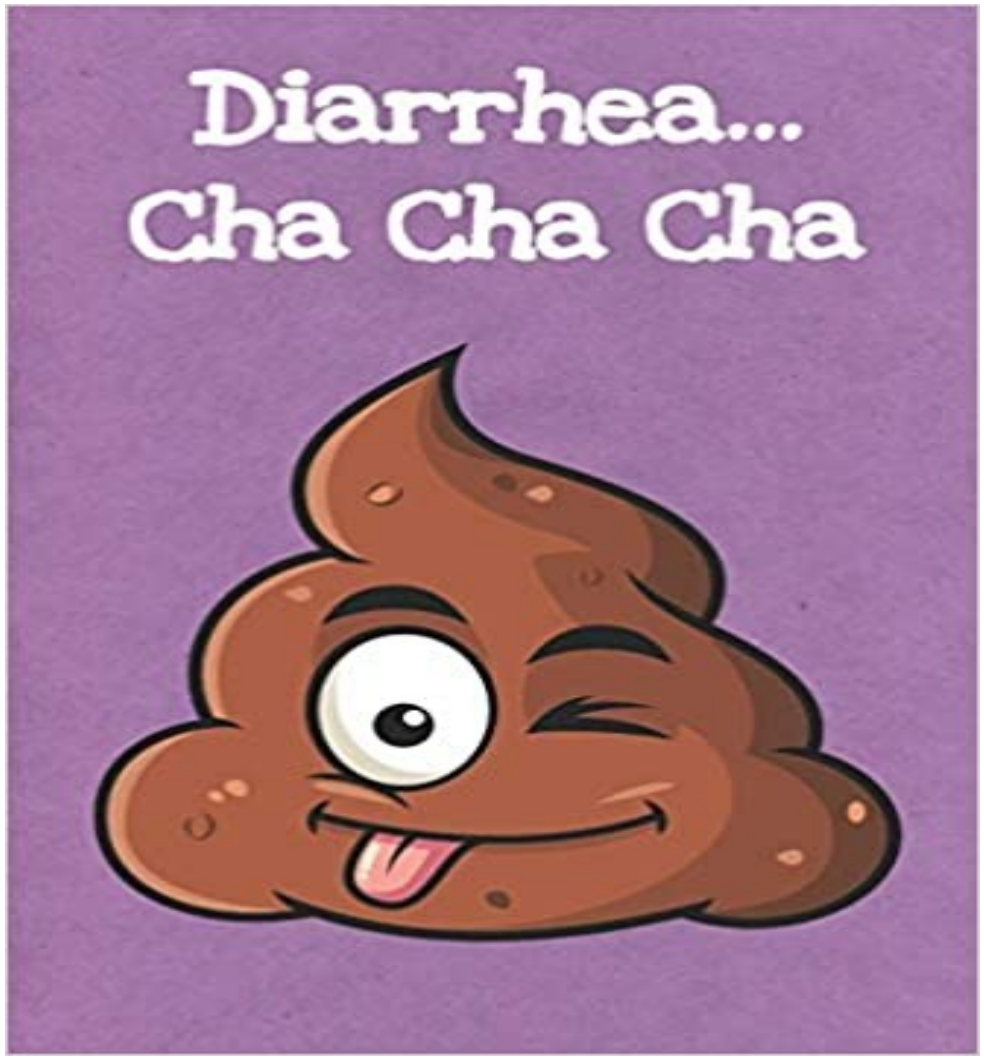
LAXATIVES AND ANTIDIARRHEAL

Learning Objectives

- **Identify** major pharmacological classes used to treat diarrhea and constipation.
- **Explain** their mechanisms of action and therapeutic roles.
- **Apply** pharmacotherapy principles to common clinical presentations.
- **Analyze** patient-specific factors affecting drug selection.
- **Evaluate** the safety and appropriateness of a pharmacotherapeutic regimen.
- **Develop** an individualized treatment plan for patients with diarrhea or constipation.



Drugs used in Diarrhea

WHO definition-

- **3 or more loose or watery stool in 24hrs period**
- Major cause of morbidity and mortality worldwide especially in developing countries
- 1000 children die everyday due to diarrhoea
- Main cause of death is dehydration
- Other causes- dysentry and undernutrition

Types of Diarrhoea

Acute diarrhoea-

- Sudden onset and lasts for less than 2 weeks
- 90%are infectious

Chronic diarrhoea-

- Lasts for more than 4 weeks
- Most cases-non infectious

Persistent diarrhoea-

- Lasts between 2-4 weeks

Principles of management

- Treatment of fluid depletion, shock & acidosis
- Maintenance of nutrition
- Drug therapy



Rehydration

- Replenishment of water and electrolytes lost through dehydration
- Both I.V and Oral

I.V rehydration-

- When fluid loss is severe i.e. >10%body weight or if the patient is losing> 10ml/kg/hr
- Unable to take enough fluids due to weakness, stupor or vomiting

Oral rehydration

STANDARD FORMULA

COMPOSITION & CONCENTRATION OF STANDARD WHO-ORS :

Ingredients	Composition (gms./L)	Ingredients	Concentration (mmol/L)
Sodium chloride	3.5	Sodium	90
Potassium chloride	1.5	Potassium	80
Trisodium citrate (anhyd.)	2.9	Citrate	10
Glucose (anhyd.)	20	Glucose	111
		Osmolarity =	311

NEW FORMULA

Table 3:
WHO/UNICEF ORS

Ingredient	g/L
Sodium chloride	2.6
Potassium chloride	1.5
Glucose, anhydrous	13.5
Trisodium citrate dihydrate	2.9
Total	20.5

Table 4:
ORS Osmolalities

Particle	mmol/L
Sodium	75
Potassium	20
Glucose	75
Chloride	65
Citrate	10
Total	245

Oral Rehydration

WHO ORS Formulae - comparison

(All values in mmol / litre)

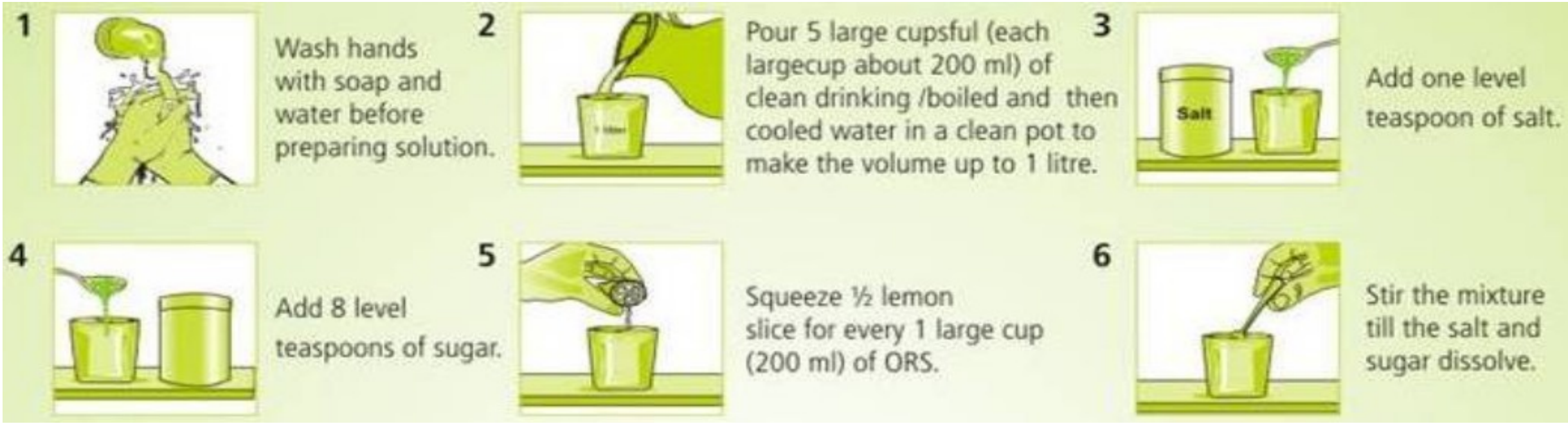
	Standard ORS (g/L)	Standard ORS (mEq/L)	Reduced osmolarity ORS (2003)
Sodium Chloride	2.6	90	75
Chloride		80	65
Potassium chloride	1.5	20	20
Trisodium Citrate	2.9	10	10
Anhydrous Glucose	13.5	111	75
Total Osmolarity		311	245

- WHO standard formula was based on cholera stools in which loss was Na+ was more
- Nowadays, majority cause of diarrhoea is Rota Virus
- In new formula, concentration of NaCl and glucose as well as total osmolarity is decreased which revealed stool volume was reduced by 20% , faster water absorption, precludes risk of hypernatremia and is cheaper



Home-made ORS solution

- Inexpensive
- Consists of 8 teaspoon of sugar and 1 teaspoon of salt mixed in 1 litre boiled and cooled drinking water



Non-diarrhoeal uses of ORT

(a) Postsurgical, postburn and post-trauma maintenance of hydration and nutrition (in place of i.v. infusion).

(b) Heat stroke.

(c) During changeover from intravenous to oral

- **Zinc in pediatric diarrhea:**
- *A*dministration of Zinc along with low osmolarity ORS reduces the duration and severity of acute diarrhoea episodes in children <5yr
- Continued Zinc supplementation

  -20 mg/day for 6–60 months age;

  -10 mg/day for 0–6 month age for 10–14 days



following the episode reduces recurrences of diarrhoea

Nutrition:

- Feeding during diarrhoea has been shown to increase intestinal digestive enzymes and cell proliferation in mucosa.

- Simple foods like breast milk or ½ strength buffalo milk, boiled potato, rice, chicken soup, banana, sago, etc. should be given as soon as the patient can eat.



Drug Therapy

- *Specific antimicrobial drugs*
- *Probiotics*
- *Drugs for IBD*
  - 5-ASA compounds
  - Corticosteroids
  - Immunosuppressants
  - TNFα inhibitors

- *Non specific antidiarrhoeal drugs*
- *Antisecretory*
  - Racecadotril
  - Bismuth subsalicylate
  - Anticholinergics
  - Octreotides
- *Antimotility-*
  - Codeine
  - Diphenoxylate
  - Loperamide

Antimicrobials are useful only in severe disease (but not in mild cases):

- ***Traveller's diarrhea****-mostly due to ETEC*

Drugs- Cotrimoxazole, norfloxacin, doxycycline reduces the duration of the diarrhea

- ***EPEC –***

Drugs- Cotrimoxazole/fluoroquinolone

- ***Shigella enteritis*** *only when associated with blood/mucus in stool*

Drugs- Ciprofloxacin/norfloxacin

- ***Nonthyphoid salmonella enteritis:***

Drugs- FQ, cotrimoxazole or ampicillin

- ***Yersinia enterocolitica***

Cotrimoxazole suitable drug in severe cases; ciprofloxacin is an alternative



Probiotics

- Are microbial cell preparation, either live culture or lypophillised powder - provides health benefits by improving/restoring gut flora
- Antibiotic use are associated with alteration in the population, composition and balance of gut microflora
- Recolonization of the gut by nonpathogenic, mostly lactic acid forming bacteria and yeast restore the balance

- Organisms most commonly used are—
- ***Lactobacillus sp.,Bifidobacterium, Streptococcus faecalis, Enterococcus sp. and the yeast Saccharomyces boulardii,*** *etc*
- Natural curd/yogurt can serve as probiotic
- Probiotics significantly reduce antibiotic-associated diarrhoea, acute infective diarrhoea and risk of traveller’s diarrhoea

Anti-secretory



Racecadotril

- new antidiarrhoeal drug
- Prodrug- forms active metabolite
- enkephalinase inhibitor that increases the local concentration of enkephalins in intestinal mucosa which then stimulates **μ** and **δ opioid receptor** and reduces intestinal secretion
- Can be given in children
- Side effects - nausea, constipation, headache
- Secreted through breast milk, hence avoided in lactating mothers

Octreotide



- Synthetic somatostatin analogue
- Has long plasma life life(1.5- 2 hours)
- More potent than somatostatin

MOA

- Inhibition of hormonal secretion
- Inhibition of release of 5-HT, gastrin, secretin, CCK, motilin and pancreatic polypeptide
- Decreases intestinal and pancreatic secretion

Reduces GI motility and inhibits gall bladder contraction

Dose-100 microgram BD or TDS subcutaneously

Uses

- Secretory diarrhoeas associated with carcinoid tumors and VIP secreting tumors
- Oesophageal varices
- Acromegaly
- Tumours(carcinoid tumour and VIP secreting tumour)

Side effects-

Short term therapy- nausea, abdominal discomfort, pain at the site of injection

*Steatorrhea (can lead to fat soluble vitamin deficiency)

Long term therapy- Acute cholecystitis, hypothyroidism

Anticholinergics

- Hyoscyamine and dicyclomine
- Decreases bowel motility which results in increased fluid absorption from the intestinal tract and decreases abdominal cramps

Bismuth subsalicylate

It has antisecretory, antibacterial and anti-inflammatory action

Side effects:

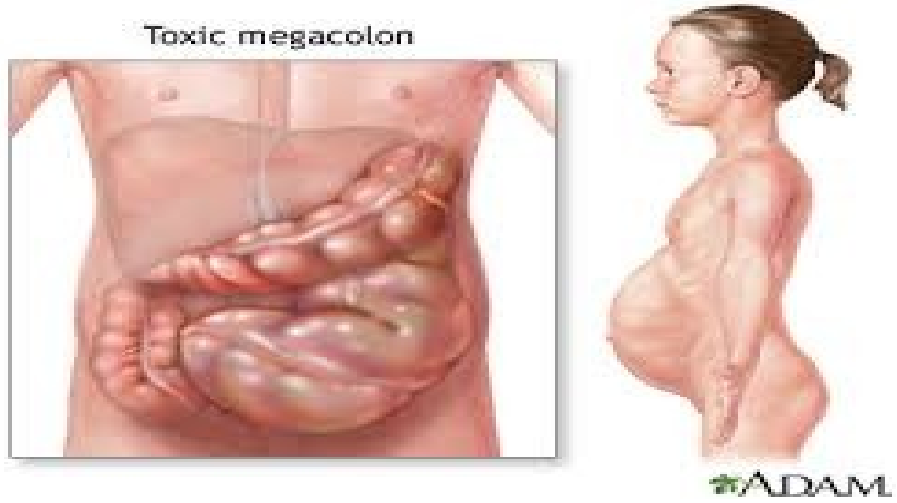
Blackening of stool
Darkening of tongue

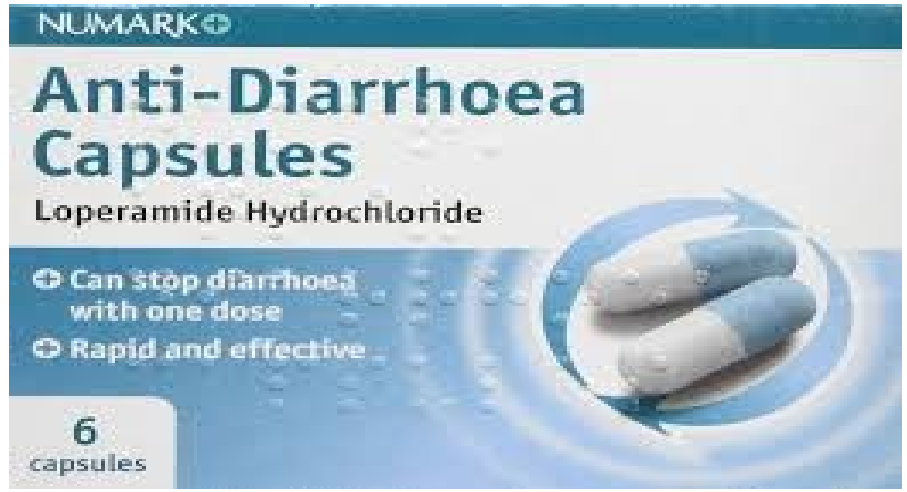
Antimotility-Loperamide

- Opiod agonist
- 40 times more potent than morphine
- Does not cross BBB, has neither analgesic effects nor abuse liability
- Inhibits secretion and motility; improves faecal incontinence by enhancing anal sphincter tone

Adverse Effects:

- Abdominal cramps, rashes, paralytic ileus
- Toxic megacolon with abdominal distension- a serious complication in young children- fatalities due to absorption of toxins from the intestines





- The use is limited to noninfective diarrhoea, mild traveller’s diarrhoea, and when diarrhoea is exhausting or idiopathic diarrhoea in AIDS patients
- They are contraindicated in acute infective diarrhoeas(*Shigella,* EPEC) because they delay clearance of the pathogen from the intestine.
- Loperamide is contraindicated in children < 4 yr.

Diphenoxylate

- less potent
- Crosses BBB
- Higher doses causes CNS depression
- Exhibit dependence liability on prolonged use
- Combined with atropine- to prevent abuse

Loperamide

- More potent
- Doesnot cross BBB
- No CNS depression
- No dependence liability
- Does not need atropine

LOMOTIL(Diphenoxylate 2.5mg and Atropine 0.5mg)

Drugs for Constipation



Constipation

- Delayed passage of faeces through the intestine(usually fewer than three times a week)

It is a symptom, not a disease

Painful bowel movements

Bloating

sensation of incomplete bowel evacuation

Causes of constipation

- Eating too less fibrous food
- Not drinking enough fluids
- Lack of exercise/ physical activity
- Change in routine
- Older age
- Frequent use of laxatives
- Certain disease and conditions
- Medications

Prevention of Constipation

- Increase in the fibrous content in the daily diet
- Increase in daily fluid intake
- Increase in physical activity
- Not neglecting the nature’s call
- Regular toilet routine
- Use drugs which causes lesser constipation as a side effects
- Correcting the underlying pathology

Pharmacological management of constipation

- **Laxatives –** milder action, elimination of soft but formed stool
- **Purgatives -** stronger action, watery evacuation

Laxatives

- Bulk forming laxatives
- Osmotic laxatives
- Lubricant laxatives
- Surfactant laxatives

Purgatives

- Osmotic purgatives
- Irritant purgatives

Laxatives are used:

- To treat constipation
- To avoid undue straining during defecation in cases having hernia, haeorrhoids
- Before or after surgery of any anorectal disease
- In bedridden patients

Bulk forming agents

Dietary fibre:

- Bran
- Psyllium
- Ispaghula
- Methyl cellulose

Dietary fibre : Bran

MOA-

Absorbs water in the intestine, swells, increases the water content of the faeces- soften it and facilitate colonic transit

Should be mixed in a glass of water/juice, stirred and drink immediately- insufficient fluid result in intestinal obstruction

Psyllium &Ispaghula

- Forms a gelatinous mass by absorbing the water and softens the stool
- Acts in 1-3 days
- If taken dry, can cause esophageal impaction
- **Dose-** 3-8 gram refined husk freshly mixed with cold milk, fruit juice or water once or twice daily

Stool Softener

Docusates (DOSS: Diocetyl sodium sulfosuccinate)

- Anionic detergents, softens the stool by decreasing the surface tension of the faecal mass to allow the water to penetrate into the stool
- Commonly recommended for patients who should avoid straining while defecating



- **Dose-** 100-400mg/day
- **Drawback**- can disrupt the mucosal barrier and enhance absorption of many non-absorbable drugs- eg: liquid paraffin- should not be combined
- **Side effects-** bitter in taste-cause nausea, Cramps and abdominal pain can occur

Liquid Paraffin

- Pharmacologically inert viseous liquid which contains mineral oil
- Dose- 15- 30 ml/day at bed time



Side effects-

- unpleasant to swallow
- Deficiency of fat soluble vitamins on chronic use- carried away with stool

Purgatives are used for complete colonic cleansing:

- Prior to gastrointestinal endoscopic procedure
- To flush out worms after the use of antihelminthic drug
- To prepare the bowel before surgery or abdominal X-ray

Osmotic Purgatives

Saline purgatives

- Magnesium sulfate
- Magnesium hydroxide
- Sodium sulfate
- Sodium phosphate
- sodium pot. Tartrate





- Soluble inorganic salts which increases the faecal bulk by retaining water by osmotic effect, thus increasing peristalsis

Contraindication

- Mag. Salts contraindicated in renal insufficiency
- Sod. Salts should not be given to patients of CHF
- Repeated use of saline puragatives results in fluid and electrolyte imbalance

Lactulose

- Semisynthetic disaccharide of fructose and lactose- neither digested nor absorbed in small intestine
- Further, broken down by colonic bacteria to osmotically more active products - draws the water into the lumen and distend it



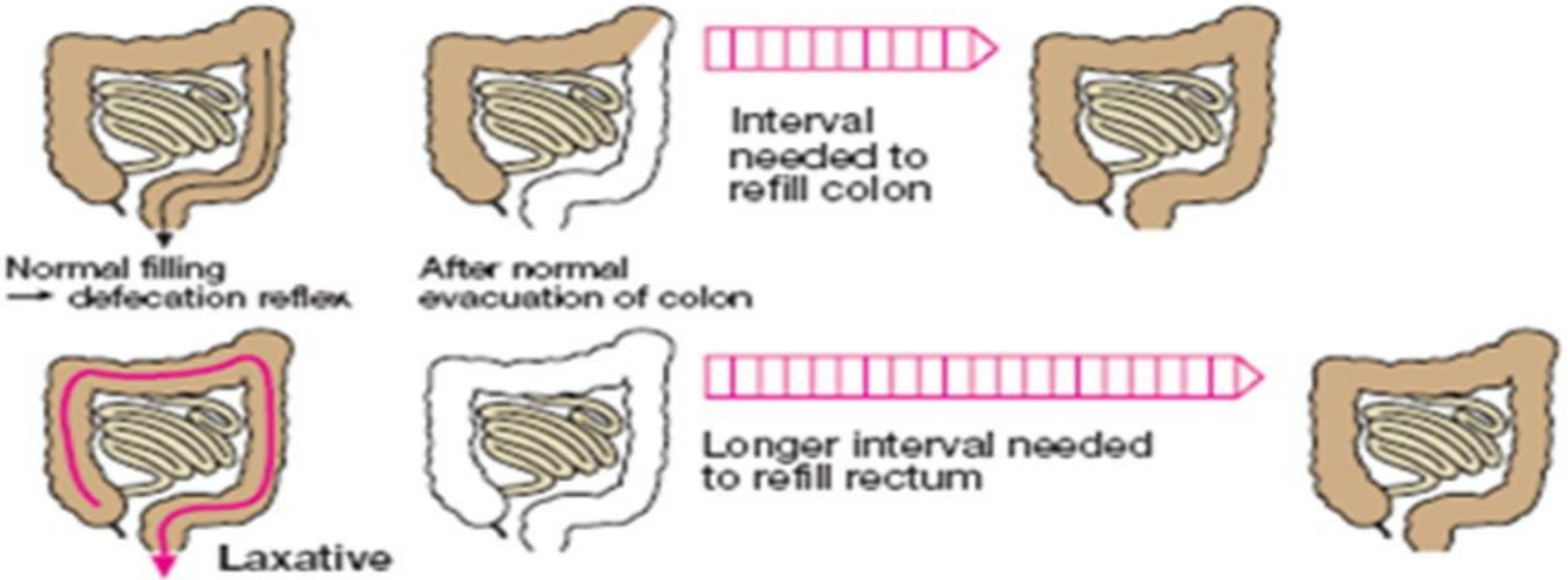
- **Dose-** 10-30 grams taken with plenty of water to produce 2 to 3 soft stools per day
- **Side effects-** flatulence, cramps
- Nausea due to its peculiar sweet taste

*Also used in the treatment of **hepatic encephalopathy** (reduce blood ammonia levels)

Laxative abuse syndrome

Mechanism:

1. With the use of strong purgatives, the colon may be so thoroughly evacuated that a bowel movement may not occur normally until a few days later. This delay reinforces the need for more laxative. Eventually the patient may require daily laxatives to maintain bowel function.



Stimulant/Irritant purgatives

Organic irritants- Bisacodyl, Sod. Picosulfate

Anthraquinones- Senna, Cascara sagrada

Oils- Castor oil







Senna, Cascara, Sagrada

- In the colon, the bacteria degrade them to active form anthrol, which either acts locally or is absorbed into the circulation
- **Dose-** 12- 25 mg at bed time
- **Side effects-**
- Secreted through breast milk- avoided in lactating mothers
- Melanosis coli- on chronic use
- Yellowish brown or red color urine

Bisacodyl

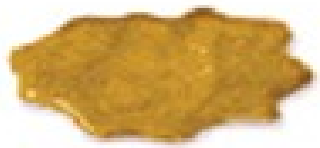
- Activated in intestine by deacetylation
- Primary site of action is colon
- Irritate the mucosa, produces mild inflamation and stimulates peristalsis
- **Dose-** 5 to 10mg at bedtime
- **Adverse effects-**
- Usually safe except for skin rashes and occasional abdominal cramps

Castor oil



- Hydrolysed in small intestine by pancreatic lipase to ricinolic acid which increases the intestinal motility
- **Dose-** 15 to 25 ml in the morning
- **Adverse effects-**
- Frequent cramping
- Stimulates uterine contraction- spontaneous abortion

BRISTOL STOOL CHART

	Type 1	Separate hard lumps	**SEVERE CONSTIPATION**
	Type 2	Lumpy and sausage like	**MILD CONSTIPATION**
	Type 3	A sausage shape with cracks in the surface	**NORMAL**
	Type 4	Like a smooth, soft sausage or snake	**NORMAL**
	Type 5	Soft blobs with clear-cut edges	**LACKING FIBRE**
	Type 6	Mushy consistency with ragged edges	**MILD DIARRHEA**
	Type 7	Liquid consistency with no solid pieces	**SEVERE DIARRHEA**

Inflammatory Bowel Disease

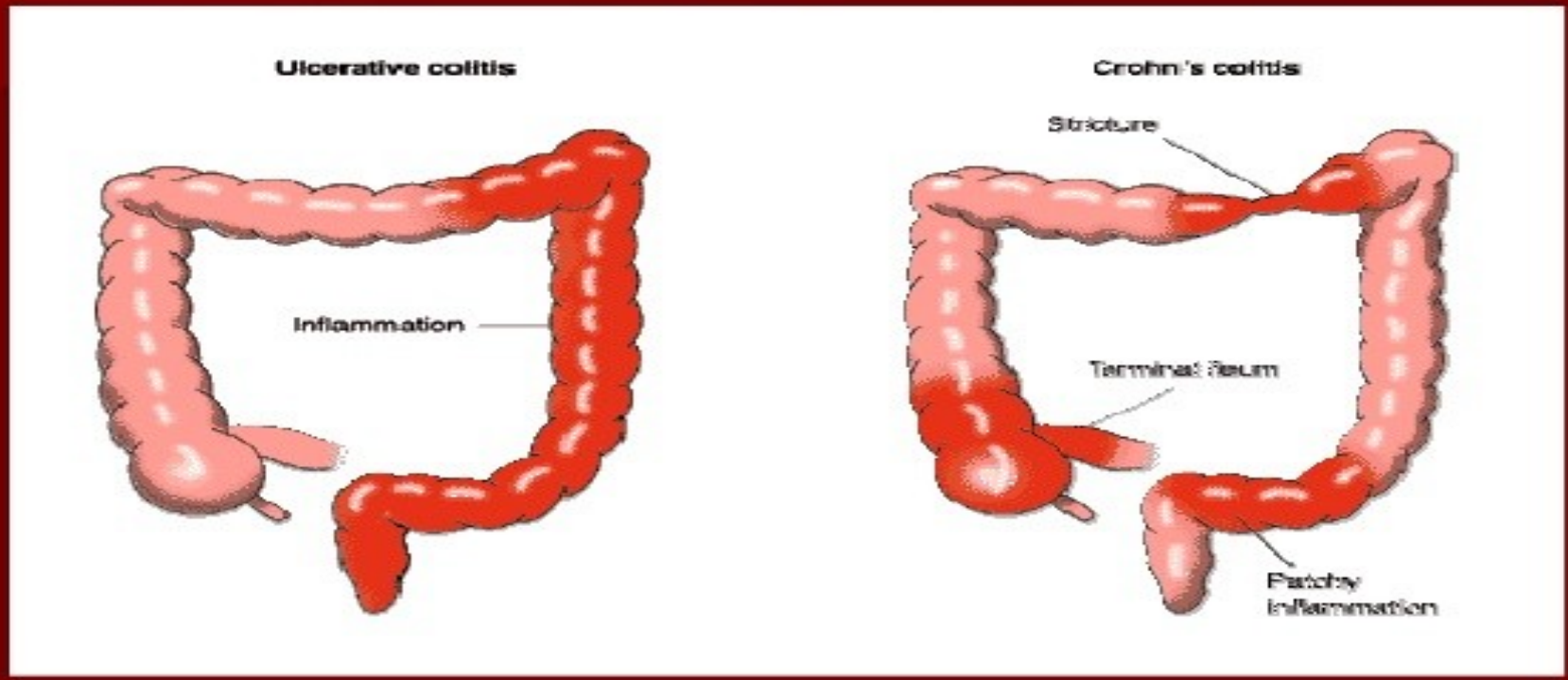
Group of chronic disorders that cause inflammation or ulceration in large and small intestines

Types

Crohn's disease(CD)- chronic granulomatous inflammation involving any location of the GIT from mouth to anus

Ulcerative colitis(UC)- non granulomatous inflammatory disorder that affects the rectum and extends proximally to affect variable extent of the colon

Inflammatory Bowel Disease



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Drugs for IBD

- *5-Amino salicylic acid (5-ASA) compounds*
- *Corticosteroids*
- *Immunosuppressants*
- *TNFα inhibitors*

5-ASA compounds

- Main anti-inflammatory compound that acts locally in the colon
- When given orally, more than 80% absorbed in proximal intestine- very few reaches to the colon
- Addition of inert compound to 5-ASA- to decrease absorption
- **Sufasalazine(5-ASA+ sulfapyridine)**
- **Olsalazine(5-ASA+5-ASA)**
- **Balsalazine(5-ASA+amino benzyl alanine)**

Sulfasalazine



- Contains 5-ASA and sulfapyridine
- Splits to release 5-ASA and sulfapyridine
- Sulfapyridine released from sulfasalazine along with 5-ASA is absorbed, undergoes acetylation and excreted in urine
- Acts **locally inhibiting the production of inflammatory mediators**
- A/E- nausea, vomiting, allergic reactions, hepatitis, pancreatitis

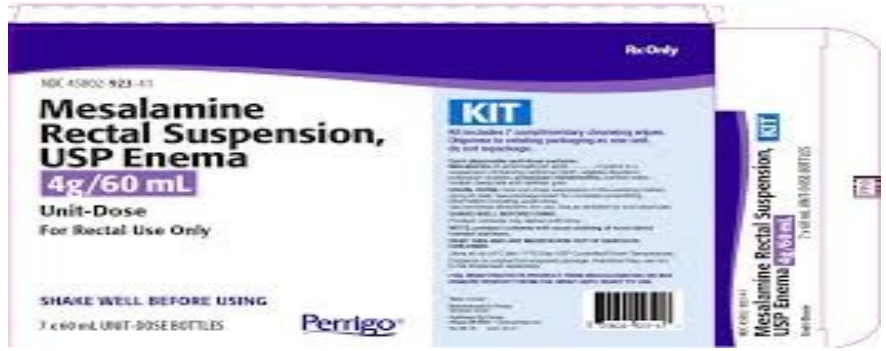
Mesalazine(mesalamine)

- Generic formulation of 5-ASA
- Primary use is in preventing relapse of UC.
- Mesalamine with pH sensitive coating- which releases 5-ASA directly in terminal ileum and colon- **more efficacious with less side effects**
- **Sulfasalazine and mesalamine- used to treat mild to moderate disease (ulcerative colitis and chron's diseases)**
- **A/E:**

- **Dose related-** headache, fatigue, nausea
- **Allergic reaction-** rash, steven johnson syndrome, hepatitis, bonemarrow supression
- **Decrease in sperm number and motility-** reversible
- **Nephrotoxicity-** risk of interstitial nephritis, hence renal function should be monitored

5-ASA enema:

- Another mode of delivery of 5-ASA to colon is to administer it by a retention enema
- useful in patients with UC confined to rectum or distal colon



Olsalazine

- Two molecules of 5-ASA coupled together by azo bond
- No separate carrier moiety is needed
- Poorly absorbed after oral administration

Balsalazine

- 5-ASA linked to inert 4-aminobenzoyl-β-alanine
- The 5-ASA is released in the colon, and the carrier is poorly absorbed
- It can be used as a safer alternative to sulfasalazine

Corticosteroids

- Prednisolone (40–60 mg/day) or equivalent are highly effective in controlling symptoms as well as in inducing remission in **both UC and CrD**

- Drugs of choice for severe exacerbations of inflammatory bowel disease

- Used for short term and tapered the doses once remission has been attained

- A sizeable percentage of severe IBD patients either relapse on stoppage of the steroids (steroid dependent) or do not respond to it (steroid resistant).
- Specific immunosuppressant drugs are strongly indicated in such IBD patients.

Immunosuppressants

- Because of **long latency of response, they are not suitable for acute flareups of the disease**, but have good remission maintaining and steroid sparing property

Azathioprine:

- Commonly used immunosuppressant in IBD
- patients may experience higher bone marrow toxicity

6-Mercaptopurine can be used in its place. It is indicated in steroid-dependent, streroid-resistant and relatively severe cases of IBD, or those who experience frequent flareups

Methotrexate

- Dihydrofolate reductase inhibitor with immunosuppressant property is a 2nd line drug in IBD, especially CrD
- **It acts faster than azathioprine and has remission inducing property as well**

Adverse effects-

- GI toxicity- nausea, vomiting , diarrhoea, stomatitis
- Hepatotoxicity
- Leucopenia



Cyclosporine

Uses-

Severe ulcerative colitis that has failed to respond adequately to steroids

To treat fistulas in Crohn’s disease

Adverse effects-

Hypertension, gingival hyperplasia, nephrotoxicity, seizures, oppurtunistic infections



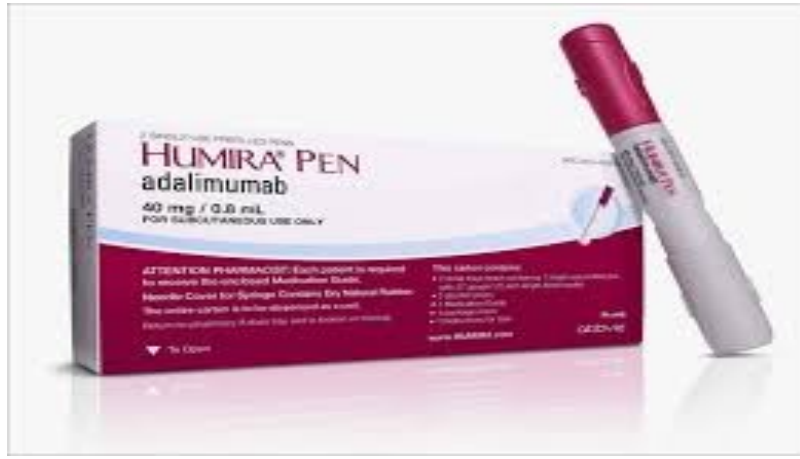
TNFα inhibitor

Infliximab-



- Chimeric mouse-human monoclonal antibody
- It binds with free as well as membrane bound TNF- prevents the cytokine from binding to its cell surface receptor
- indicated in severe active CrD, fistulating CrD and severe UC not improved with I.V corticosteroids and immunosuppressant
- It is only a reserve drug for selected patients with refractory disease

- **Adalimumab**
- Recombinant human anti-TNF alpha monoclonal antibody
- Block the inflammatory activity of TNF-alpha




Thank You